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import requests, json, os, time

OVERPASS = 'https://overpass-api.de/api/interpreter'

# 江戸川区の駅のみ（大島・西大島・東大島は江東区のため除外）
STATIONS = [
    {'id': 'nishikasai', 'name': '西葛西', 'line': '東京メトロ東西線', 'fl': 4},
    {'id': 'kasai', 'name': '葛西', 'line': '東京メトロ東西線', 'fl': 3},
    {'id': 'kasairinkai', 'name': '葛西臨海公園', 'line': 'JR京葉線', 'fl': 5},
    {'id': 'funabori', 'name': '船堀', 'line': '都営新宿線', 'fl': 3},
    {'id': 'ichinoe', 'name': '一之江', 'line': '都営新宿線', 'fl': 3},
    {'id': 'mizue', 'name': '瑞江', 'line': '都営新宿線', 'fl': 2},
    {'id': 'shinozaki', 'name': '篠崎', 'line': '都営新宿線', 'fl': 2},
    {'id': 'koiwa', 'name': '小岩', 'line': 'JR総武線', 'fl': 2},
    {'id': 'shinkoiwa', 'name': '新小岩', 'line': 'JR総武線', 'fl': 2},
    {'id': 'hirai', 'name': '平井', 'line': 'JR総武線', 'fl': 1},
]

RADIUS = 1000

# 江戸川区のバウンディングボックス（緯度35.63～35.75、経度139.82～139.95）
EDOGAWA_BBOX = '35.63,139.82,35.75,139.95'

def get_station_coord(name):
    q = f'''
[out:json][timeout:30];
node["railway"="station"]["name"="{name}"]
({EDOGAWA_BBOX});
out 1;
'''
    try:
        r = requests.post(OVERPASS, data={'data': q}, timeout=30)
        els = r.json().get('elements', [])
        if els:
            return els[0]['lat'], els[0]['lon']
    except Exception as e:
        print(f' 座標取得エラー {name}: {e}')
    return None, None

def get_facilities(lat, lng):
    q = f'''
[out:json][timeout:45];
(
    node["leisure"="park"](around:{RADIUS},{lat},{lng});

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way['leisure'='park'](around:{RADIUS},{lat},{lng});
node['shop'='supermarket'](around:{RADIUS},{lat},{lng});
node['amenity'='veterinary'](around:{RADIUS},{lat},{lng});
node['amenity'='cafe']['dog'='yes'](around:{RADIUS},{lat},{lng});
node['leisure'='dog_park'](around:{RADIUS},{lat},{lng});
node['shop'='pet'](around:{RADIUS},{lat},{lng});
node['amenity'='car_sharing'](around:{RADIUS},{lat},{lng});
);
out center;
'''

try:
    r = requests.post(OVERPASS, data={'data': q}, timeout=45)
    els = r.json().get('elements', [])
    parks, shops, vets, cafes, dogruns, pets, carshares = [], [], [], [], [], []
    for el in els:
        tags = el.get('tags', {})
        name = tags.get('name', '')
        if not name:
            continue
        if tags.get('leisure') == 'park':           parks.append(name)
        elif tags.get('leisure') == 'dog_park':      dogruns.append(name)
        elif tags.get('shop') == 'supermarket':      shops.append(name)
        elif tags.get('amenity') == 'veterinary':    vets.append(name)
        elif tags.get('amenity') == 'cafe':          cafes.append(name)
        elif tags.get('shop') == 'pet':              pets.append(name)
        elif tags.get('amenity') == 'car_sharing':   carshares.append(name)
    return {
        'parks':    parks[:5],
        'shops':    shops[:5],
        'vets':     vets[:5],
        'cafes':    cafes[:3],
        'dogruns':  dogruns[:3],
        'pets':     pets[:3],
        'carshares':carshares[:3],
    }
except Exception as e:
    print(f'  施設取得エラー: {e}')
    return {'parks':[], 'shops':[], 'vets':[], 'cafes':[], 'dogruns':[], 'pets':[]}

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統計推定値 (Phase2で実データ化予定)

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HOUSING = {
    'nishikasai':62, 'kasai':63, 'kasairinkai':50,
    'funabori':66,   'ichinoe':68, 'mizue':72,
    'shinozaki':76,  'koiwa':70,   'shinkoiwa':68, 'hirai':70,
}

SAFETY = {
    'nishikasai':78, 'kasai':74, 'kasairinkai':82,

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        'funabori':72,    'ichinoe':74, 'mizue':76,
        'shinozaki':82,  'koiwa':55,   'shinkoiwa':54, 'hirai':70,
    }
}

COMMUNITY = {
    'nishikasai':71, 'kasai':69, 'kasairinkai':65,
    'funabori':67,   'ichinoe':69, 'mizue':70,
    'shinozaki':79,  'koiwa':70,   'shinkoiwa':68, 'hirai':72,
}

def main():
    print('🐕 INUMAP データ収集開始（江戸川区限定）')
    results = []
    for st in STATIONS:
        print(f'\n📍 {st['name']}駅...')
        lat, lng = get_station_coord(st['name'])
        if lat is None:
            print(f'⚠️ 座標取得失敗、スキップ')
            continue
        print(f' 座標: {lat}, {lng}')
        time.sleep(1)
        fac = get_facilities(lat, lng)
        print(f' 公園:{len(fac['parks'])} 店:{len(fac['shops'])} 動物病院:{len(fac['vets'])}')
        time.sleep(1.5)

        walk      = min(100, 40 + len(fac['parks']) * 8 + len(fac['dogruns']) * 15)
        medical   = min(100, 30 + len(fac['vets']) * 12 + len(fac['cafes']) * 8 +
        mobility  = min(100, 50 + len(fac['carshares']) * 10)

    results.append({
        'id':      st['id'],
        'name':    st['name'],
        'line':    st['line'],
        'lat':     lat,
        'lng':     lng,
        'walk':    walk,
        'housing': HOUSING.get(st['id'], 65),
        'medical': medical,
        'mobility': mobility,
        'community': COMMUNITY.get(st['id'], 68),
        'safety':  SAFETY.get(st['id'], 70),
        'fl':      st['fl'],
        'parks':   fac['parks'],
        'shops':   fac['shops'],
        'vets':    fac['vets'],
        'cafes':   fac['cafes'],
        'dogruns': fac['dogruns'],
        'carshares': fac['carshares'],
    })

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    ))

    os.makedirs('data', exist_ok=True)
    with open('data/stations.json', 'w', encoding='utf-8') as f:
        json.dump(results, f, ensure_ascii=False, indent=2)
    print(f'\n✅ {len(results)} 駅保存完了 (江戸川区) ')

if __name__ == '__main__':
    main()
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